Experiment No. 06: Exploring Recurrence Relations in Computational Algorithms.

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Date: 04/03/2026 ¶

Example 1: Write a recurrence relation for the number of ways to arrange n distinct objects in a row.\n Ans: Recurrence relation is $Y_n = nY_{(n-1)}$

In [41]:
```
from sympy import Function, rsolve
var('n')
y = Function('y')
f = y(n) - n*y(n-1)
b = rsolve(f, y(n), {y(1): 1})
b
```

Out[41]: $n!$

Example 2:An elf has a staircase of n stairs to climb. Each step it takes can cover either one stair or two stairs. Find a recurrence relation for the number of different ways for the elf to climb the n- stair staircase and also solve thar recurrence relation. Ans: Recurrence relation is $Y_{(n)} = Y_{(n-1)} + Y_{(n-2)}$

In [42]:
```
from sympy import Function, rsolve
var('n')
y = Function('y')
f = y(n)-y(n-1) - y(n-2)
b = rsolve(f, y(n), {y(1): 1, y(2): 2})
b
```

Out[42]:
$$\left(\frac{1}{2} - \frac{\sqrt{5}}{2}\right)^{n} \left(\frac{1}{2} - \frac{\sqrt{5}}{10}\right) + \left(\frac{1}{2} + \frac{\sqrt{5}}{2}\right)^{n} \left(\frac{\sqrt{5}}{10} + \frac{1}{2}\right)$$

Example 3:The Tower of Hanoi is a game consisting of n circular rings of varying size and three pegs on which the rings fit. Intially all the rings are placed on the one peg with larget ring at the bottom covered by successively smaller rings. The rules of puzzle allow disks to be moved on at a time from one peg to another as long as a disk is never placed on top of a smaller disk. The goal of puzzle is to have all the disks on second pegin order of size, with the largest on bottom. Ans: Recurrence relation is $Y_{(n)} = 2Y_{(n-1)} + 1$

In [43]:
```
from sympy import Function, rsolve
var('n')
y = Function('y')
f = y(n)-2*y(n-1)-1
b = rsolve(f, y(n), {y(1): 1})
b
```

Out[43]: $2^{n} - 1$

Example 4: Suppose we draw n straight lines on a piece of paper so that every pair of lines intersect (but no three lines intersect at a common point). Into how many regions do these n lines divide the plane? Ans: Recurrence relation is $Y_{(n)} = Y_{(n-1)} + n$

In [44]:
```
from sympy import Function, rsolve
var('n')
y = Function('y')
f = y(n)-y(n-1)-n
b = rsolve(f, y(n), {y(0): 1})
b
```

Out[44]:
$$\frac{n\left(n+1\right)}{2} + 1$$

Example 5:A bank pays 8 percent intrest each year on money in saving accounts. Find the recurrence relations for the amount of money a man would have after n years if it follows the investment strategies of: (a) Investing Rs 1000 and leaving it in the bank for n years. (b) Investing Rs 100 at the end of each year Ans: (a) Recurrence relation is $Y_{(n)} = 1.08Y_{(n-1)}$ (b) Recurrence relation is $Y_{(n)} = 1.08Y_{(n-1)} + 100$

In [45]:
```
from sympy import Function, rsolve
var('n')
y = Function('y')
f = y(n)-1.08*y(n-1)
b = rsolve(f, y(n), {y(0): 1000})
b
```

Out[45]: $1000 \cdot 1.08^n$

In [46]:
```
from sympy import Function, rsolve
var('n')
y = Function('y')
f = y(n)-1.08*y(n-1)-100
b = rsolve(f, y(n), {y(0): 1000})
b
```

Out[46]: $2250.0 \cdot 1.08^n - 1250.0$

Example 6: Solve $y_{n+2} = 5y_{n+1} - 7n$ with the intial conditions $y_0 = 1, y_1 = 1$.

In [47]:
```
from sympy import Function, rsolve
var('n')
y = Function('y')
f = y(n+2) - 5*y(n+1) + 7*n
b = rsolve(f, y(n), {y(0): 1, y(1): 1})
b
```

Exercise

1. Example Solve $a_n = a_{n-1} + 2a_{n-2}$, with the intial conditions $a_0 = 0,\ a_1 = 1$.

In [48]:
```
from sympy import Function, rsolve
var('n')
a = Function('a')
f = a(n)-a(n-1)+2*a(n-2)
b = rsolve(f, a(n), {a(0): 0, a(1):1})
b
```

Out[48]:

$$\frac{\sqrt{7}i\left(\frac{1}{2} - \frac{\sqrt{7}i}{2}\right)^n}{7} - \frac{\sqrt{7}i\left(\frac{1}{2} + \frac{\sqrt{7}i}{2}\right)^n}{7}$$

1. Example Solve $a_{n+2} = 5a_{n+1} + 6a_n$, with the intial conditions $a_0 = 0,\ a_1 = -1$.

In [49]:
```
from sympy import Function, rsolve
var('n')
a = Function('a')
f = a(n+2)-5*a(n+1)+6*a(n)
b = rsolve(f, a(n), {a(0): 0, a(1):-1})
b
```

Out[49]: $2^n - 3^n$

1. Example Solve $a_{n+2} = a_{n+1} + a_n + n^2$, with the intial conditions $a_0 = 0,\ a_1 = 1$.

In [50]:
```
from sympy import Function, rsolve
var('n')
a = Function('a')
f = a(n+2)-a(n+1)-a(n)-(n*n)
b = rsolve(f, a(n), {a(0): 0, a(1):1})
b
```

Out[50]:
$$n^2\left(-\frac{\sqrt{5}}{4}-\frac{1}{4}\right)+\left(\frac{1}{2}-\frac{\sqrt{5}}{2}\right)^n\left(-\frac{\sqrt{5}}{4}-\frac{1}{4}\right)+\left(\frac{1}{4}+\frac{\sqrt{5}}{4}\right)\left(\frac{1}{2}+\frac{\sqrt{5}}{2}\right)^n$$

1. Find the number of binary sequences of length having no 11 by using recurrence relation. Ans: $a_n = a_{n-1} + a_{n-2}$

In [51]:
```
from sympy import Function, rsolve
var('n')
a = Function('a')
f = a(n)-a(n-1)-a(n-2)
b = rsolve(f, a(n), {a(1):2,a(2):3})
b
```

Out[51]:
$$\left(\frac{1}{2}-\frac{\sqrt{5}}{2}\right)^n\left(\frac{1}{2}-\frac{3\sqrt{5}}{10}\right)+\left(\frac{1}{2}+\frac{3\sqrt{5}}{10}\right)\left(\frac{1}{2}+\frac{\sqrt{5}}{2}\right)^n$$

1. Every year Dr Finch's rabbit population doubles. He started with six rabbits. How many rabbits does he have after eight years? After n years? Ans: $a_n = 2a_{n-1}$

In [52]:
```
from sympy import Function, rsolve
var('n')
a = Function('a')
f = a(n)-2*a(n-1)
b = rsolve(f, a(n), {a(0):6})
b
```

Out[52]: $6 \cdot 2^n$

Learning:(Write in details about your Learnings from the activity):

This experiment helped me see how recurrence relations aren’t just abstract math but a way to break big problems into smaller steps. Whether it was rabbits doubling, climbing stairs, or saving money, each situation could be modeled with a simple recurrence and solved with SageMath. I realized that recurrence relations are everywhere in algorithms and daily life, and using code made it easier to understand and visualize their growth.